## Research activity

**The point of a line closest to a given point — 35 min.**

In an orthonormal frame of space, consider $A(1;0;0)$, $M(0;2;3)$ and the line $d$ through $A$ with direction vector $\vec{u} = (1;1;0)$. We look for the point of $d$ closest to $M$.

1. Express, in terms of a real parameter $t$, the coordinates of a point $H$ of $d$.
2. Find the value of $t$ for which $\overrightarrow{MH} \cdot \vec{u} = 0$. Deduce the coordinates of the resulting point $H$.
3. Calculate the distance $MH$.
4. Let $P$ be any point of $d$. Prove that $MP^2 = MH^2 + HP^2$, then deduce that $H$ is the unique point of $d$ closest to $M$.